TECHNOLOGY BRIEFING

Flink DAG vs ADK Graph

Two maps of steps, built for two different jobs

What is a DAG?

A flowchart that a computer can run, step by step.

Boxes

Each box is a step of work.

Arrows

Arrows show what must finish before what.

No loops back

The flow only moves forward, so it is predictable.

Flink: the assembly line

Processes a nonstop flow of small events at massive scale.

Moves

Payments, clicks and sensor readings, millions per second.

Speed

Each step takes milliseconds.

Guarantee

No data lost, none counted twice, even after failures.

ADK graph: the case desk

Controls how AI agents handle one task at a time.

AI where needed

Plain code does routine steps. AI is called only for language reasoning.

Human approval

Any step can pause for a person and resume later.

Audit trail

Every step is recorded, with predictable routing.

How they differ

	Flink DAG	ADK graph
Analogy	Assembly line	Case-handling desk
Flows	Nonstop events	One task at a time
Step speed	Milliseconds	Seconds (AI calls)
Shape	Fixed once deployed	Can branch and loop
Strength	Volume and speed	Controlled AI decisions

Better together

Flink
Filters and enriches the event firehose.

→

ADK graph
Runs AI reasoning and approvals on each work item.

ADK 2.0 is new and its behavior at large scale is not yet documented publicly. Proposed next step: a small pilot before committing.